

DeWang Li

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Summary

Distributed systems Senior engineer focused on large-scale AI infrastructure, high-performance storage, and LLM training systems. Experienced in latency-critical optimization, NUMA-aware scheduling, and DeepSpeed-based training workflows. Strong background in optimizing real-world systems under high concurrency and large-scale data workloads.

Experience

JD.com

2025 – 2026

Distributed Systems Senior Engineer (AI Image Infrastructure)

- Worked on a distributed image system serving **1B images/day** with **300B+ total images stored**
- Led architecture redesign replacing HTTP pipeline (Tomcat → Nginx) with **JNI-based direct invocation**
- Eliminated inter-process network and serialization overhead across services
- Achieved **40% reduction in TP99 latency** and **35% reduction in average latency** (production traffic)
- Improved system throughput and stability under high-concurrency workloads
- Optimized image transcoding pipeline (PNG → AVIF), addressing compression inefficiencies and color issues

Huawei Technologies

2022 – 2024

Distributed Systems Senior Engineer (AI Storage)

- Optimized distributed storage system for AI workloads and high-throughput data access
- Led read path redesign with metadata prefetch during open phase
- Improved throughput from **120K → 180K OPS (+50%)**
- Implemented NUMA-aware scheduling and CPU binding strategy
- Achieved **+5 GB/s bandwidth improvement per node**
- Improved system efficiency under large-scale training data pipelines

Open Source & LLM Infrastructure

DeepSpeed Contributor (Microsoft)

- Contributed multiple PRs to DeepSpeed for large-scale model training and inference systems
- Fixed correctness issue in **pass@k evaluation** for repeated prompts (LLM benchmarking reliability)
- Implemented best-checkpoint saving based on evaluation metrics
- Improved training stability, evaluation consistency, and workflow usability

Selected Contributions

- DeepSpeed #7908, #7928, #7763
- OpenRLHF #1206 — Tinker #172, #286

Technical Skills

Languages: Python, C++, Bash

Systems: Distributed Systems, NUMA Optimization, High-performance Storage, Concurrency Control

AI Infra: PyTorch, DeepSpeed, LLM Training

Performance: Latency Optimization, Throughput Scaling, System Bottleneck Analysis

Tools: Linux, Docker, Git

Education

University of Electronic Science and Technology of China

2019 – 2022

M.S. in Computer Science

Edge AI Systems (Academic Project)

- Built distributed edge AI systems using **Kubernetes** and **KubeEdge**
- Managed multi-node edge environments and deployment workflows
- Explored AI model deployment and orchestration in resource-constrained environments

Key Strengths

- System-level optimization with real production impact (latency ↓40%, throughput ↑50%)
- Strong ownership in performance-critical paths (I/O, NUMA, RPC removal)
- Bridging distributed systems and LLM infrastructure
- Experience working on large-scale, latency-sensitive production systems